$X \to Y$
$X \mapsto Y$
$F \Rightarrow G$
$F \circ G$
$x \in X$
$A \subseteq B$
$a \neq b$
$V \otimes W$
$V \oplus W$
$X \sqcup Y$
$X \sqcap Y$
$X \uplus Y$
$A \cap B$
$A \cup B$
$\bigcup A_n$
$\bigcap A_n$
$\prod$
$\lfloor x \rfloor$
$\lceil x \rceil$

ΔY
$\sum_n$
$\varnothing$
$\in$
$\subseteq$
χ
τ
ε
η

Set theory

Power set A^X

2^X

Power set A^X is a generalization of the **cartesian product A^n = A x ... x A**

A^x can be thought of as a **product of A indexed by X**

i.e. all possible X-indexed **combinations $(a_x)_{x \in X}$**, or equivalently, all **functions** $X \to A$

$A^n = A \times A \times \dots \times A$
$= $ all combination $(a_1, a_2, \dots, a_n)$

Generalizing to any set X

$A^X = \prod_{x} A$
$= $ set of all combinations $(a_x)_{x \in X}$
$= \{ f \mid f: X \to A \}$ with $f(x) = a_x$

power set 2^X is a special case that represents the set of all subsets of X

$2^X = \{ f \mid f: X \to \{0,1\} \}$ $x \in$ subset S iff $f(x) = 1$
$= \{ S: S \subseteq X \}$

Equivalence

Equivalence relation ~

Definition:

~ $\subseteq$ **X x X** is an equivalence relation if:

x ~ x for all x **reflexive**

x ~ y <=> y ~ x **symmetry**

x ~ y, y ~ z => x ~ z **transitive**

Equivalence class [x] ⊆ X is a **partition** of X by equivalence class:

[x] = { y ∈ X: y ~ x }

Quotient X/~ is

X/~ = { [x]: x ∈ X }

Proposition (partition):
Equivalence classes **partition** X.
i.e.

[x] ∩ [y] =	[x]	if x ~ y
	∅	otherwise

Equivalence relation f~g
Let f, g: X → Y
Let **f ~ g** be the equivalence relation **induced** by

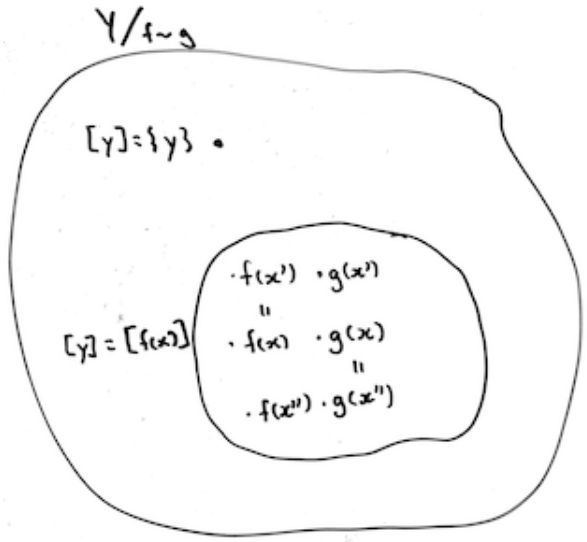
f(x) ~ g(x)

Proposition
In f~g

x ~ y where x ≠ y <=> there is a sequence (xi, yi)

(xi, yi) = (f(x), g(x)) or (g(x), f(x))

x = **x0 ~ y0 = x1 ~ y1 = x2 ~ y2 ... ~ yn** = y



Geometry

Rotation matrix
If Rθ is the rotation matrix by θ degrees in 2 dimensions, then

Rθ(1, 0) = (cos θ, sin θ)
Rθ(0, 1) = (-sin θ, cos θ)

Hence

Rθ **= cos** θ **-sin** θ
sin θ **cos** θ

For any point (cos φ, sin φ) on the circle

Rθ(cos φ, sin φ) = (cos θ cos φ - sin θ sin φ, sin θ cos φ + cos θ sin φ)

= R(θ + φ) (1, 0)

Hence the trigonometric identities

cos (θ **+** φ**)** **= cos** θ **cos** φ **- sin** θ **sin** φ
sin (θ **+** φ**)** **= sin** θ **cos** φ **+ cos** θ **sin** φ

and when φ = θ

cos 2θ **= cos^2** θ **- sin^2** θ
sin 2θ **= 2 sin** θ **cos** θ

(cos - λ)^2 + sin^2 = λ^2 - 2λcos + 1
= (λ - e^iθ) (λ + e^-iθ)

x cos - y sin = kx
y sin + x cos = ky

x (cos - k) - y sin = 0
y (sin -k) + x cos = 0

Tensor
Tensor product v ⊗ w (of vectors)
tensor product: https://en.wikipedia.org/wiki/Tensor_product
Tensor product ⊗ is a generalization of **multiplication**.
The main characteristic of multiplication is that it is **bilinear** (**multilinear**)

multiplication
ab
product ab
(a + b) (c + d) = ac + ad + bc + bd bilinear

AB = { ab: a ∈ A, b ∈ B }
product space AB
· : A x B → AB
product operation

tensor
v⊗w **tensor**
v⊗w
(v + v')⊗(w+w') = v⊗w + v⊗w' + v'⊗w + v'⊗w'
bilinear

V ⊗ W = { v ⊗ w : v ∈ V, w ∈ W } **tensor**

product space V ⊗ W (Vector space)
⊗: V x W → V ⊗ W **tensor product operation (bilinear)**

Tensor product vector space V ⊗ W
Tensor product ⊗: V x W → V ⊗ W (bilinear operation)
tensor product: https://en.wikipedia.org/wiki/Tensor_product
If V, W are vector spaces, then **V ⊗ W** is a vector space

Definition
Let vi, wj be basis of V, W
V ⊗ W is a space generated with basis

vi⊗wj

Hence

vector v⊗w = ∑_i,j (ai bj) · vi⊗wj v = ∑ ai vi w = ∑ bj wj

addition v⊗w + v'⊗w' = ∑_i,j (ai bj + ai' b'j) · vi⊗wj
multiplication c (v⊗w) = ∑_i,j (c ai bj) · vi⊗wj

Tensor algebra T(V)
https://en.wikipedia.org/wiki/Tensor_product
Tensor algebra T(V) is essentially a **free** extension of the **tensor product** ⊗.
Put another way it is a **free algebra** of a **vector space V**.
i.e. the most **general algebra** constructable from a vector space V.
In particular, it contains all possible tensor products

(v1⊗ ... ⊗vk) ⊗ (w1⊗...⊗wl) = v1⊗ ... ⊗vk⊗w1⊗...⊗wl
∈ ⊗V^k ⊗V^l ∈ ⊗V^(k+l)

Definition: Let

⊗V^k = V ⊗ V ... ⊗ V k products

Tensor algebra T(V) of a vector space V is defined by

T(V) = ⊕_k ⊗V^k

= K ⊗V^0 scalar
⊕ V ⊗V^1 vector
⊕ ⊗V^2 multiple of 2 vectors
⊕ ⊗V^3 multiple of 3 vectors
...

i.e. T(V) is a **free sum** of all tensors

k + v + v1⊗v2 + v3⊗v4⊗v5 + ... ∈ T(V) element where only finite terms are ≠ 0

Multiplication is generated by the **canonical tensor product.**
Let

⊗: **⊗V^k x ⊗V^l → ⊗V^k+l**
(v1⊗...⊗vk) ⊗ (w1⊗...⊗wl) = v1⊗...⊗vk⊗w1⊗...⊗wl

i.e. any tensors can be multiplied
e.g.

V		**⊗V^2**		**⊗V^3**
s	⊗	u⊗v	=	s⊗u⊗w

⊗V^2		**⊗V^3**		**⊗V^5**
s⊗t	⊗	u⊗v⊗w	=	t⊗s⊗u⊗v⊗w

Multiplication of T(V) is simply a **free, bilinear** application of the tensor product

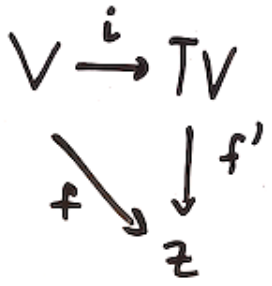
⊗: **T(V) x T(V) → T(V)**
(k + v + v1⊗v2 + v3⊗v4⊗v5 + ...) ⊗ (l + w + w1⊗w2 + w3⊗w4⊗w5 + ...)

= kl + kw + k(w1⊗w2) + k(w3⊗w4⊗w5) + ...
+ lv + v⊗w + v⊗w1⊗w2 + v⊗w3⊗w4⊗w5 + ...
... etc ...

Tensor algebra is a **universal property**

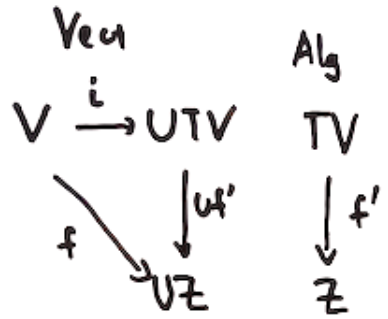
Vector space		**Algebra**
i.e. linear map		i.e. bilinear map
f: V → Z	<=>	f': TV → Z
	f = f'	

where **i: V → TV injection**.



In category theory terms:

Free functor	T: Vec → Alg		
Forgetful functor	U: Alg → Vec		
injection as **universal** map	i: V → UTV		
Adjoint	{ f: V → UZ }	<=>	{ f': TV → Z }
	Vec	f = f' i	**Alg**



V x W (cartesian product) vs V ⊕ W (direct sum) vs V ⊗ W (tensor)
Different types of **linearity**

linearity

V x W

element (v,w)

V⊕W: Generalization of **addition**.
Addition, but V and W are **disjoint**
i.e.

v+w = (0 + w) + (v + 0)
c(v+w) = cv + cw

(v+v')+(w+w') = (v+w) + (v'+w')

element v⊕w

as **free object** formal **sum**
(∑civi) + (∑diwi) basis {vi} of V
basis {wi} of w

linear
scale c(v⊕w) = cv⊕cw
addition c(v+w) = cv + cw
additive (v+v')⊕(w+w') = v⊕w+v'⊕w'
(v+v')+(w+w') = (v+w) + (v'+w)
= v⊕w'+v'⊕w
= (v+v')⊕w'+0⊕w
= (v+v')⊕w+0⊕w'

= v⊕(w+w')+v'⊕0
= v'⊕(w+w')+v⊕0

linear map Let f: V→R, g: W→S be linear maps.
Then

f⊕g:V⊕W → R⊕S
f⊕g(v⊕w) = f(v)⊕g(w)

f⊕g(c(v⊕w)) = c (f(v)⊕g(w))

f⊕g((v+v')⊕(w+w')) = f(v)⊕g(w) + f(v')⊕g(w')

= f(v')⊕g(w) + f(v)⊕g(w')

= ...

V⊗W: Generalization of **multiplication**.
https://en.wikipedia.org/wiki/Tensor_product
i.e.

c(vw) = (cv)w = v(cw)

(v+v')(w+w') = vw + vw' + v'w + vw'
= v(w+w') + v'(w+w')
...

Let **{vi}** and **{dj}** be a **basis** for V and W respectively.
Let e ∈ {ei} d ∈ {dj}

element vi⊗wj ∈ V⊗W

multilinear

scale c(vi⊗wj) = c vi⊗wj
= vi⊗cwj

additive (vi+vi')⊗(wj+wj') = (vi+vi')⊗wj + vi⊗(+wj')
= vi⊗wj + vi⊗wj' + vi'⊗wj + vi'⊗wj'

Hence for any v = ∑ vi*(v) vi ∈ V, w = ∑ wi*(w) w ∈ W
note: vi*: V → F is the **dual** aka coefficient of vi

element v⊗w = ∑i,j vi*(v) wj*(w) (vi⊗wi) i.e. every combination (vi⊗wi)

linear map Let f: V→R, g: W→S be linear maps.
Then

f⊗g: V⊗W → R⊗S
f⊗g(vi⊗wj) = f(v)⊗g(wj)

scale f⊗g(c(vi⊗wj)) = c (f(vi)⊗g(wj))

linear f⊗g((vi+vi')⊗wj) = f(vi)⊗g(wj) +f(vi')⊗g(wj)

f⊗g(vi⊗(wj+wj')) = f(vi)⊗g(wj) +f(vi)⊗g(wj')

Hence, for any v ∈ V, w ∈ W

f⊗g(v⊗w)) = ∑ vi*(v) wj*(w) f(vi) ⊗ g(wj)

= f(v)⊗g(w)

V∧V: alternating

element	$v \wedge w$	
alternating		
scale	$c(v \wedge w)$	$= cv \wedge w$
additive	$(v+v') \wedge (w+w')$	$= v \wedge w + v \wedge w' + v' \wedge w + v' \wedge w'$
alternating	$v \wedge v$	$= 0$
	hence anticommutative	
	$v \wedge w$	$= - w \wedge v$

Exterior algebra, wedge product

Wedge product v $\wedge$ w

https://en.wikipedia.org/wiki/Exterior_algebra

The **wedge product v** $\wedge$ **w** is a **free alternating, bilinear form**

i.e. an extension of the tensor product $v \otimes w$

alternating	$v \wedge v = 0$
bilinear	$(v + v') \wedge (w+w') = (v \wedge w) + (v \wedge w') + (v' \wedge w) + (v' \wedge w')$

A consequence is anticommutativity

anticommutative	$v \wedge w = -w \wedge v$	since $0 = (v+w) \wedge (v+w)$ $= (v \wedge w) + (w \wedge v)$

It can be interpreted as a generalization of the **determinant**

bilinear	det (..., u+v, ...)	= det (..., u, ...) + det (..., v, ...)
alternating	det (..., u ..., u, ...)	= 0

If A = {aij} is an nxn matrix with column vectors vj

$a_j = \sum a_{ij} e_i$

Then

det A	$= \sum_{\sigma \in Sn} \operatorname{sgn}(\sigma) \prod a_{i,\sigma(i)}$
a1 $\wedge$ **...** $\wedge$ **an**	$= \sum_{\sigma \in Sn} \operatorname{sgn}(\sigma) \prod a_{i,\sigma(i)} (e1 \wedge ... \wedge en)$

Put another way, it a free of the **inclusion-exclusion principle** via the **anticommutativity** $v \wedge w = - w \wedge v$.

e.g. if v1...v3 are a basis

a $\wedge$ **b** $\wedge$ **c**	=	$(a1v1+a2v2+a3v3) \wedge (b1v1+b2v2+b3v3) \wedge (c1v1+c2v2+c3v3)$
	=	$a1b2c3\ (v1 \wedge v2 \wedge v3) + a1b3c2\ (v1 \wedge v3 \wedge v2)$ $+ a2b1c3\ (v2 \wedge v1 \wedge v3) + a2b3c1\ (v2 \wedge v3 \wedge v1)$

+ a3b1c2 (v3∧v1∧v2) + a3b2c1 (v3∧v2∧v1)

= a1b2c3 (v1∧v2∧v3) - a1b3c2 (v1∧v2∧v3)
anticommutativity vi ∧ vj = -vj ∧ vi
- a2b1c3 (v1∧v2∧v3) + a2b3c1 (v1∧v2∧v3)
+ a3b1c2 (v1∧v2∧v3) - a3b2c1 (v1∧v2∧v3)

= **(a1b2c3 - a1b3c2 - a2b1c3 + a2b3c1 + a3b1c2 - a3b2c1) v1∧v2∧v3**

Exterior algebra ∧V
Exterior algebra ∧V is the **free algebra** of the **wedge product v1 ∧ ... ∧ vn.**
i.e. a free combination of all possible wedge products

v1∧...∧vk ∧ (w1∧...∧wl) = v1∧...∧vk∧w1∧...∧wl

Formally, it is defined as the **tensor product T(V)** modulo all possible **alternations v⊗v**

∧V = T(V) mod I I = { v⊗v : v ∈ T(V) }
i.e. alternations

Multiplication is the **free wedge product v∧w**

∧**:** ∧**V** → ∧**V**
v∧w = v⊗w (mod I)

Let e1, ..., en be a basis of V and v1, ..., vn be any vectors of V with

vj = ∑i aij ei

Then

v1∧...vn **=** ∑ **(**a1,σ(i) eσ(1) ∧ ... ∧ an,σ(n) eσ(n))
σ∈Sn

= ∑ sgn σ ∏ai,σ(i) **(**e1∧ ...∧ en)
σ∈Sn

= **(det A) e1∧ ... en** A = { aij }

For any vectors v1, ..., vn and permutation σ ∈ Sn

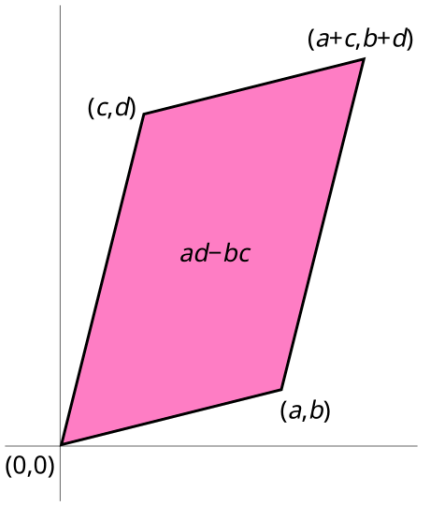
σ(1) ∧ ... ∧ vσ(n) = sgn σ^-1 (v1 ∧ .... ∧ vn) note: since
= sgn σ (v1 ∧ .... ∧ vn) - each flip of kth and k+1th position is a **transpose (k k+1)**
- product of transposes is σ**^-1**

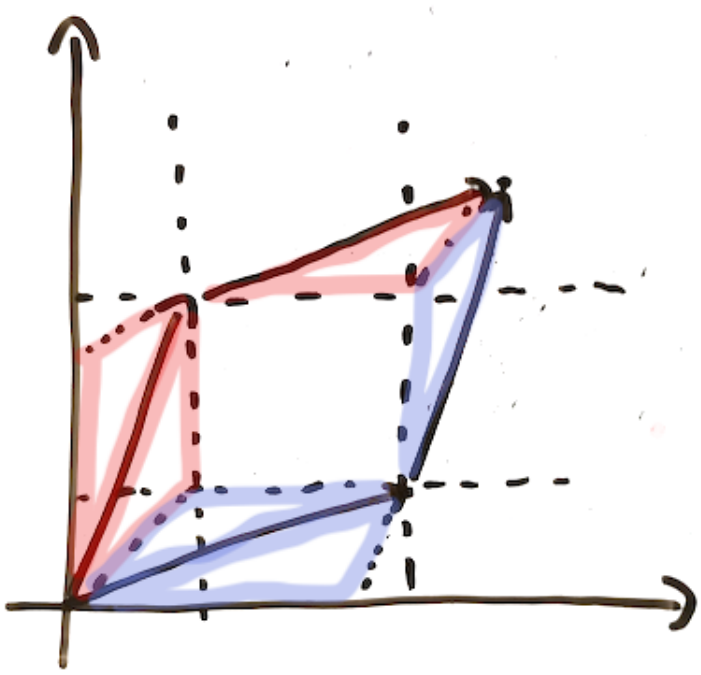
-- Interpretation --
Exterior algebra is a generalization of the **area**
i.e. computing products by the **inclusion-exclusion** principle.

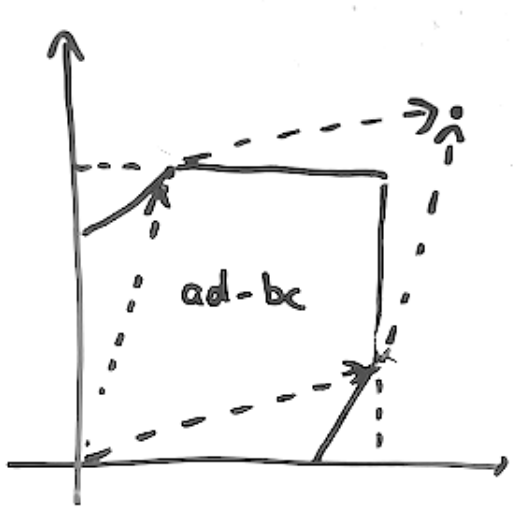
e.g. in R ∧ R
Let

v1 = (a, b)
= ae1+ ce2

v2 = (c, d)
= be1+ de2







Then

v1 + v2 = (a+c, b+d)

v1∧ v2 = ac(e1∧e1) + ad(e1∧e2) + bc(e2∧e1) + bd(e2∧e2)

= **(ad-bc) e1∧e2**

Ring

Group ring

Group ring: https://en.wikipedia.org/wiki/Group_ring

A form of **free algebra** generalizing the **algebra of polynomials**

Just replace **variables** with group **elements**.

Polynomials R[x]

$\Sigma_n r_n x^n$ elements

$\Sigma_n r_n x^n \cdot \Sigma_n r'_n x^n = \Sigma_{n=i+j}(r_i + r'_j)x^n$ multiplication

Group ring R[G]

$\Sigma_g r_g[g]$ elements

$\Sigma_g r_g[g] \cdot \Sigma_g r'_g[g] = \Sigma_{g=ij}(r_i r'_j)[g]$ multiplication

Definition

Let G be a group, and R a ring.

The **group ring R[G]** is an R-algebra with elements of the form

$\Sigma_g r_g[g]$ where r_g ≠ 0 for finite g

[g] = free value i.e. formal

where

r (∑_g r_g [g]) = ∑_g (r · r_g) [g] scalar multiplication by R

$\sum_g r_g [g] + \sum_g r'_g [g] = \sum_g (r_g + r'_g) [g]$ addition

$\sum_g r_g [g] \cdot \sum_g r'_g [g] = \sum_{g=ij} (r_i \; r_j) [g]$
multiplication

Special values

$r_g = 0$ all	0	zero (+ identity)
$r_g = 1$ g=1 0 otherwise	[1]	one (* identity)
$r_g = 1$ g=g0 0 otherwise	[g0]	group element g0
$r_g = r0$ g=1 0 otherwise	r0 [1]	ring element r0

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==== Algebraic Structures ====

Semigroup
Monoid
A semigroup is the **weakest** form of group with only **associativity**.

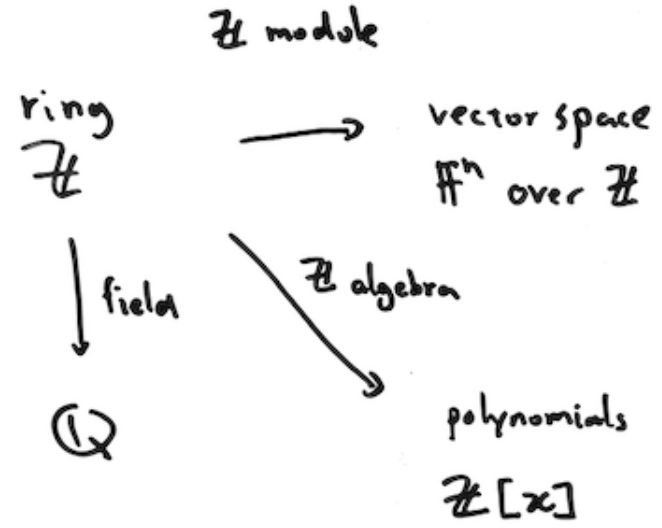
Definition:
A **semigroup** is a set S with an **associative** binary operation $\circ : S \times S \rightarrow S$

$(a \circ b) \circ c = a \circ (b \circ c)$

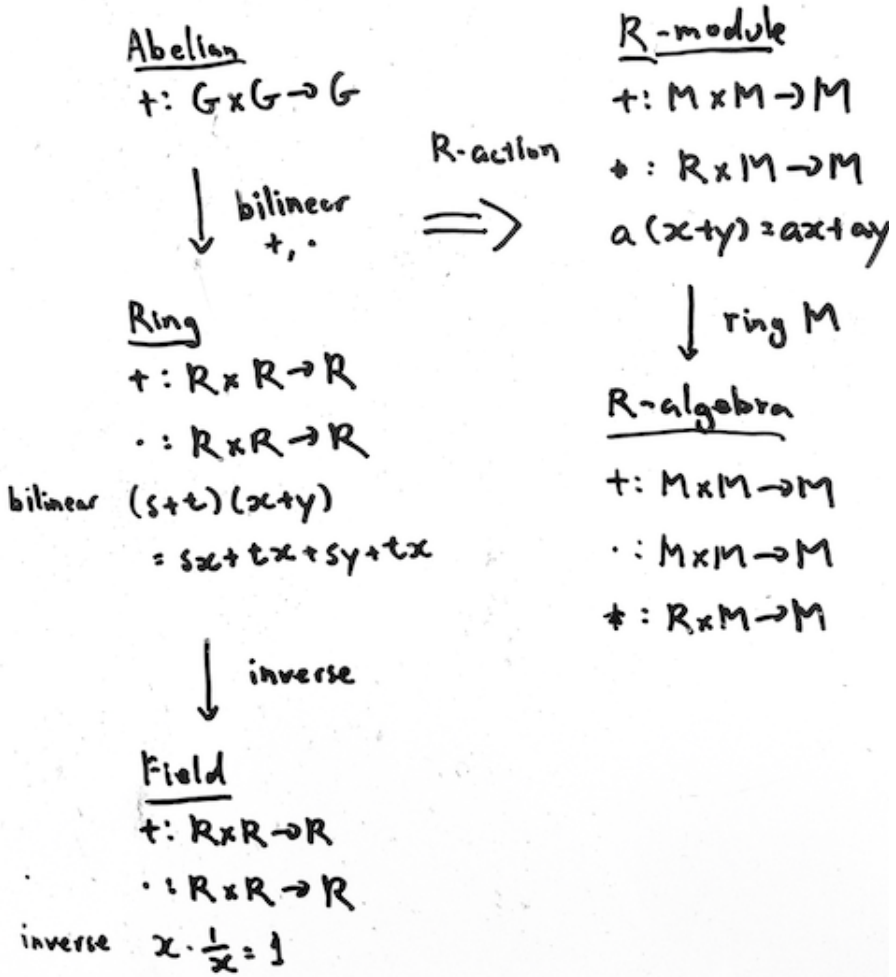
Group variations

Semigroup	associative only (no identity, inverse)
Monoid	associative + identity
Group	associative + identity + inverse
Groupoid	binary operation is **partial**

Group, ring, field
canonical



Relationships



Definition

Source

Group

Abelian group
Group + commutative +: G x G → G integer with addition

Ring
Abelian + bilinear multiplication +: R x R → R integer with addition & multiplication (no inverse required)
(not group) ・: R x R → R
(x + y)(u + v) = xu + xv + yu + yv

Field
Ring + multiplicative inverse + : K x K → K rational numbers with addition & multiplication (invertible)
i.e. * is a group operation ・: K x K → K
x ・ 1/x = 1

Modules, algebras
Algebra over a field: https://en.wikipedia.org/wiki/Algebra_over_a_field
i.e. spaces with **elements M** and **scalar R**

	Definition
Source	
R-module	
Abelian M + R scalar multiplication	+: M x M → M
vector space	
	*: R x M → M
Q x Q with scalars Z = Z-module	
	a(x+y) = ax + ay
R-algebra (ring)	
Ring M + R scalar multiplication	+: M x M → M
polynomials K[x]	(x^a + x^b)(x^c + x^d) = x^ac + x^ad + x^bc + x^bd
	・: M x M → M
complex numbers	(a + bi) (c + di) = (ac - bd) + (ad + bc)i
	*: R x M → M scalar multiplication

==========
====
==== Group ====

Theorem: Quotient
If N ⊆ G is normal, then G/N is a group
i.e.
G/N = { gN : g ∈ G }

1N ∈ G/N **identity**
gN ・ hN = (gh) N **multiplication**

==========
Proof
- gN ・ hN is well-defined:
For any n, n' ∈ N

gn ・ hn' = g(hh^-1 n) ・ hn'

= (gh) (h^-1nh)n'
∈ (gh) N since h^-1nh ∈ N and n' ∈ N

=> well-defined

QED
==========

Theorem: Universal property
Quotient is a universal construct
i.e.
Let N ⊆ G normal
There is a 1-1 relation

{ f: G → H | N ⊆ ker f } <=> { f': G/N → H }
f = f' ∘ q where q: G → G/N
f(g) = f'(gN)

==========
Proof:
=> Suppose f: G → H.
Let
f': G/N → H
f'(gN) = f(g)

claim: f' is well-defined
If n ∈ N, then f(gn) = f(g) f(n) = f(g)

<= Suppose f': G/N → H
Let
f: G → H i.e. f = f' ∘ q where q: G → G/N quotient
f(g) = f'(gN)

QED
==========

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---- OLD ----

V x W
(v, w)
(v1 + v2, w1 + w2) = -
No linearity specified

V ⊕ W
(v, w)
(v1 + v2, w1 + w2) = (v1, w1) + (v2, w2)
Coordinate-wise linear

V ⊗ W
(v, w)
(v1 + v2, w1 + w2) = (v1, w1 + w2) + (v2, w1 + w2)
Bilinear
= (v1, w1) + (v1, w2) + (v2, w1) + (v2, w2)

A way to **generate** a **K-algebra** from a **K-vector space**
In particular, a way to define **multiplication** using the vectors.

Tensor algebra is simply a **formal (free) combination** of all possible products so that **any tensors** can be multiplied
Let

⊗V^k = V ⊗ V ... ⊗ V k products

Tensor algebra T(V) of a vector space V is defined by

T(V) = ⊕_k ⊗V^k

= K ⊗V^0 scalar
⊕ V ⊗V^1 vector
⊕ ⊗V^2 multiple of 2 vectors
⊕ ⊗V^3 multiple of 3 vectors
...

i.e. T(V) is a **free** sum of tensors

k + v + v1⊗v2 + v3⊗v4⊗v5 + ... ∈ T(V) element where only finite terms are ≠ 0

Multiplication is generated **canonical tensor product.**
Let

⊗: **⊗V^k x ⊗V^l → ⊗V^k+l**
(v1⊗...⊗vk) ⊗ (w1⊗...⊗wl) = v1⊗...⊗vk⊗w1⊗...⊗wl

Any tensors can be multiplied
e.g.

V		**⊗V^2**		**⊗V^3**
s	⊗	u⊗v	=	s⊗u⊗w

⊗V^2		**⊗V^3**		**⊗V^5**
s⊗t	⊗	u⊗v⊗w	=	t⊗s⊗u⊗v⊗w

Multiplication of T(V) is simply a **bilinear extension** of this product:

⊗: **T(V) x T(V) → T(V)**
(k + v + v1⊗v2 + v3⊗v4⊗v5 + ...) ⊗ (l + w + w1⊗w2 + w3⊗w4⊗w5 + ...)

= kl + kw + k(w1⊗w2) + k(w3⊗w4⊗w5) + ...
+ lv + v⊗w + v⊗w1⊗w2 + v⊗w3⊗w4⊗w5 + ...
... etc ...

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OLD

Multiplication is **term-wise** i.e.

(k, v, v1⊗v2, v3⊗v4⊗v5, ...) ⊗ (l, w, w1⊗w2, w3⊗w4⊗w5, ...)
= (kl, kw, kw1⊗w2, kw3⊗w4⊗w5, ...)
+ (0, lv, v⊗w, v⊗w1⊗w2, v⊗w3⊗w4⊗w5, ...)

(k, v, v1⊗v2, v3⊗v4⊗v5, ...) ⊗ (l, w, w1⊗w2, w3⊗w4⊗w5, ...)

= (
kl, scalar
0,
v⊗w, vector ⊗ vector = multiple of 2
0,
v1⊗v2⊗w1⊗w2, multiple of 2 ⊗ multiple of 2 = multiple of 4 vectors
0,
0,
v3⊗v4⊗v5⊗w3⊗w4⊗w5, multiple of 3 ⊗ multiple of 3 = multiple 6 vectors
0,
...
)

Theorem (determinant)
Let A be an nxn matrix

A = { aij } elements aij i,j=1...n

= (a1 a2 ... am) column vectors aj
aj = ∑ aij ei

Then

a1 ∧ .. ∧an = (det A) e1∧...∧en

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proof
Simply by the alternation

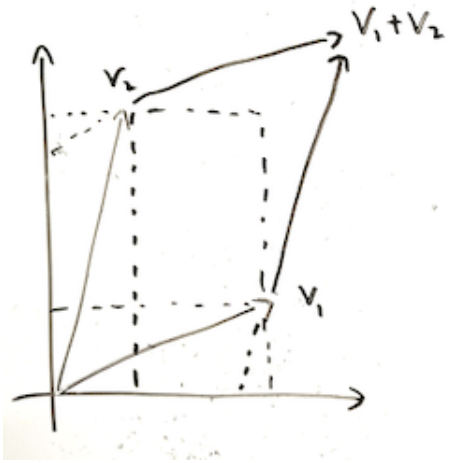
a1 ∧ ... ∧ an = (∑ai1ei) ∧ (∑aj2 ej) ∧ ...

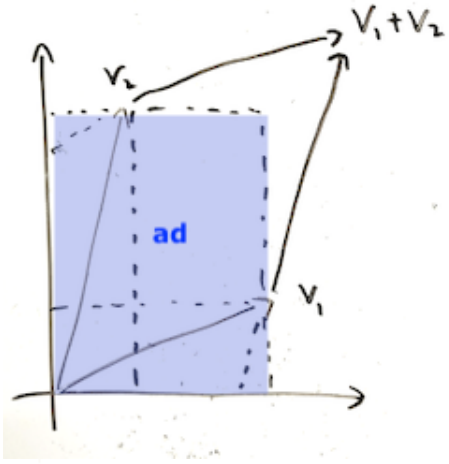
= ∑ (ai1aj2 ak3 ...) ei∧ej∧ek... sum over all combinations

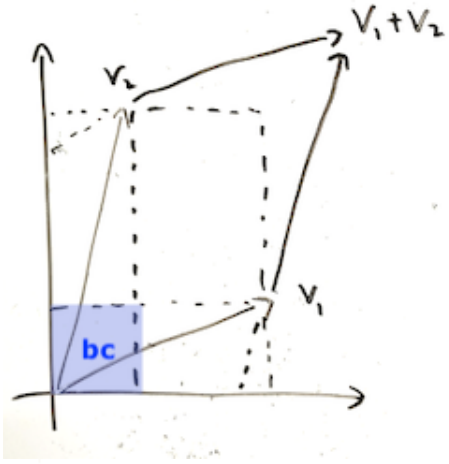
= ∑ sgn(σ) (∏ aσ(t),t) (e1∧...∧en) sum of all σ ∈ Sn since any term with the same ei vanish

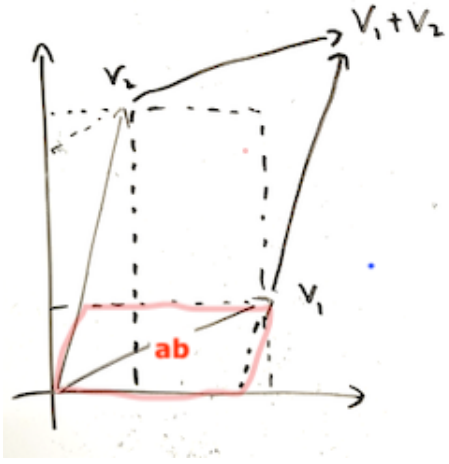
= det A (e1∧...∧en)

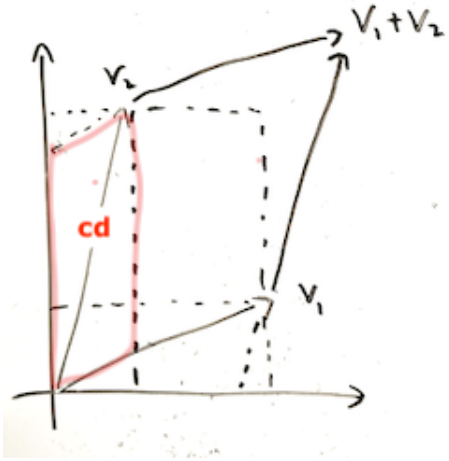
QED
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$V_1 + V_2$
V_2
V_1


$V_1 + V_2$
V_2
ad
V_1


$V_1 + V_2$
V_2
V_1
bc


$V_1 + V_2$
V_2
V_1
ab


$V_1 + V_2$
V_2
cd
V_1